

b. If x and y are real numbers and $(x + iy)(2 + i) = 3 - i$, find the values of x and y .

$$(x + yi)(2 + i) = 2x + xi + 2yi - y$$

$$2x - y + (x + 2y)i = 3 - i$$

$$2x - y = 3$$

$$y = 2x - 3$$

$$x + 2y = -1$$

$$x + 2(2x - 3) = -1$$

$$x + 4x - 6 = -1$$

$$5x = 5$$

$$x = 1$$

$$y = -1$$

c. Solve the following equation for x in terms of w .

$$2\sqrt{x-3} - w\sqrt{x} = 0$$